



Influence of reaction mechanisms based the turbulence and heat release interaction in a combustor stabilized with a V-shaped stabilizer

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ABSTRACT

Chemical reaction mechanism plays a key role in combustion simulation, especially the near blowoff condition which has intense interaction of reaction and turbulence. Therefore, five kerosene reaction mechanisms were applied in (Ma=0.1 and Ma=0.2) bluff body reacting and nonreacting flow simulation to study the influence of mechanisms. The major novelty here as compared to previous contributions is studying the role of reaction mechanisms in simulating near extinction states from the perspective of the interaction between heat release, turbulence, and chemical reactions by comparing mechanisms of different types and complexities. The results indicate that, firstly, the reflux zone, shear boundary layer, and heat release rate are strongly influenced by the type of mechanism, which further affects the flame stretching rate. Secondly, high stretching rate will weaken the fuel pyrolysis process of ARC type mechanism, reducing the concentration of free radicals and small molecule hydrocarbons, thereby disrupting the combustion reaction. Thirdly, the insufficient pyrolysis products in the reaction layer cannot support the subsequent oxidation heat release, which is the reason for the extinguishing of this type of mechanism. Fourthly, although the general contracting mechanism represented by JL4 performs poorly in terms of reaction kinetics parameters, its omission of the process of pyrolysis into small molecule hydrocarbons makes it insensitive to flame elongation, allowing it to burn stably. These findings not only provide a reference for reaction mechanism selection for numerical simulation of afterburner flame stabilizer, but set foundation of applicability research of mechanisms.

Table of Parameter symbols:

Parameter symbols	Physical meaning
ρ	Density
R	Droplet radius in Lagrange coordinates
N	Droplet number
$\mathbf{V}$	Droplet velocity
Θ	Droplet temperature
$\mathbf{X}$	Droplet position
$\bar{p}$	Droplet joint probability density function
Da	Damköhler number
D'	Diffusion coefficient
Ka	Karlovitz number
p	Pressure
τ_{sgs}	Subgrid scale
τ_{ij}^{sgs}	Subgrid stress tensor
ϕ_α	Scalar Number of components
Ψ	Set of scalars sample space
$\bar{p}_{sgs}$	Subgrid probability density function

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$\bar{S}_{mass}$	The source term of the mass equation
$\bar{S}_{mom,i}$	The source term of the momentum equation
$\bar{S}_\alpha$	The source term of the scalar transport equation
HRR	Total heat release rate
K_s	Two-dimensional Karlovitz numbers
K_{ext}	The strain rate obtained after treatment
σ_{ij}	Viscous stress tensor
u_i	Velocity in the i direction
D	Width of bluff body

1. Introduction

According to the request of the high performance aero-engine enhancing the temperature of the combustor chamber and the combustion efficiency with few pollution has become an essential goal of experts and engineers [1,2]. However, It is unfeasible to directly simulate the turbulent flame based on detailed chemical models in the

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combustion model because of the high computational complexity and numerical rigidity of the solution [3,4]. Therefore, analyzing simplified chemistry represents an attractive trade-off between precision and efficiency [5].

The current numerical simulation of combustion reactions consists of two types of reaction schemes: one is the overall reaction mechanisms, which consist of some empirical reactions [6–8]; Another type is Analytically Reduced Chemistry (ARC) schemes, which simplify by analyzing large-scale detailed mechanisms to remove redundant reactions and components [9,10].

The overall reaction fits heat release to reflect combustion characteristics. It decomposes the overall reaction of fuel oxidation into multiple intermediate steps and several key intermediates. Dozens of experts has proposed some useful reaction mechanisms: Westbrook proposed a kind of one step reaction mechanism and two step reaction mechanism [11]; Dong gave a reaction mechanism consists of four steps [12]; With further research, more and more mechanisms has been proposed, Jones brought up a four step seven component reaction mechanism with $C_{12}H_{23}$ as the average molecular formula [13]; Franzelli improved the rate parameters of the reaction according to different equivalent ratios, and obtained the total packet mechanism with a wider application range on account of the work of Westbrook [14]. The mechanisms above have been widely applied in the simulation of the combustion chamber. Whereas, too much simplification of the overall mechanisms makes the applied range of overall finite and exist the disadvantage of the lack of the information of pollutant such as CO and NO_x [15]. In addition, the complex combustion phenomenon simulation based on overall is inaccuracy due to its oversimplification of the actual reaction mechanism [16].

In view of defects above, ARC has become new research hotspot because of its sensible simplification and more detailed reaction mechanisms [17]. Chang pointed out that large hydrocarbon fuels showed significant similarities in terms of combustion characteristics, which made comparisons between different fuel molecular formulas possible [18]; Zettervall proposed a molecular formula $C_{12}H_{23}$ and reaction mechanism consisting of 65 steps irreversible reaction 22 components of Jet A-1 fuel, and based on proposed mechanisms further advance new chemical reaction mechanism consisting of 77 steps 22 components [19]; Wang et al. proposed the reduced combustion mechanism of $C_{10}H_{12}$ and $C_{12}H_{23}$ [20]; You proposed a detailed kinetic model for the normal alkanes up to n-dodecane with verified the validity of the model by experiments consisting of fuel pyrolysis in plug flow and jet-stirred reactors, laminar flame speeds, and ignition delay times behind reflected shock waves [21]; Sarathy proposed a detailed and reduced chemical kinetic mechanism for singly methylated iso-alkanes ranging from C7 to C20, which is used to show how the presence of a methyl branch affects important combustion properties [22].

The advantages of ARC containing accuracy made it widely accepted and used to simulate the complex reaction field which usually coupled with complicated flow field, such as the combustion chamber in aero-engines, and the complexity of Modern aeroengine combustion chambers, more and more experts begin studying it by computational fluid dynamics (CFD) [23]. The computing methods contains direct numerical simulation (DNS), large eddy simulations (LES) and Reynolds average simulation (RANS) [24]. DNS directly solve the basic equations of fluid dynamics based on numerical methods, which lead large amount of calculation. In contrast, Reynolds Equation was solved in RANS to get steady flow with ignored the fluctuation which is significant to the flow field of combustion chamber. Compared to above two methods, LES keep balance of accuracy and costs by filtering the Navier Stocks equation based on the filtering scale between large eddies and small eddies to separate the flow into directly solved large scale quantities and small scale quantities which need subgrid model to solve [25]. According to the advantages of LES, it has been used to simulate the combustion chamber with coupled the chemical reaction mechanism [26].

The coupling method between LES and chemical reactions varies depending on different combustion modes. In actual combustion chamber, the state of combustion is between premixed and non premixed, and some areas within the flow field blowoff and reignition. Chemical reactions are no longer infinitely fast compared to mixing processes, and the finite reaction rate effect must be considered [27]. To address this issue, the Transport Probability Density Function (TPDF) method solves the probability density function of transport variables. Therefore, compared to the infinite rate assumption, more accurate simulation of blowoff and ignition processes has become an important direction in the current numerical simulation research of turbulent combustion based on LES [28].

Due to the effectiveness of LES-TPDF, many experts have conducted research and got achievement based on it: Hodzic et al. [29] used LES-TPDF approach to study the flame behind the bluff body in near-blowoff conditions, and the statistics of the simulation results were in an agreement with the experimental values. Hasti et al. [30] used the LES-TPDF method to study the effect of reaction and diffusion time scale on the flame stabilization process of the jet-in-hot-coflow (JHC) CH_4/H_2 under simulated moderate or intense low oxygen dilution (MILD) combustion and proved that the LES-TPDF simulations yield improved predictions of the species and temperature fields. All these indicate that LES is a high-resolution method for solving turbulent flow. TPDF can couple detailed chemical reaction mechanisms without any modeling [31], which has a significant advantage in simulating phenomena where chemical reactions play a major role. Several studies have confirmed that LES-TPDF coupled with detailed chemical reaction mechanisms is a high-resolution numerical simulation method, which can simultaneously resolve turbulence and chemical reactions and can be used to study blowoff problems.

For reasons above, in this paper, the LES-TPDF coupled with a suitable chemical reaction mechanism was used to calculated the transient and time-averaged fields of complex turbulent combustion during blowoff which has intense interaction of turbulence and reaction. And since the chemical reaction kinetic model significantly impacts the CFD flow field simulation, the influence of chemical mechanism needs to be studied for the detailed case, so that providing a foundation for further research on what kind of mechanisms are applicable for blowoff simulation.

To achieve the above research objectives, the method is verified by two-phase bluff body flames with the distribution of the velocity of flow direction and turbulent kinetic energy firstly. On the basis of verifying the effectiveness of the method, the flow characters under different reaction mechanisms consisting of the reflux zone, shear layer, HRR and structure of flame have been gotten and the blow-off occurred in simulation has been analysed by species distribution, flame stretching rate, and HRR, which provides a reference for the numerical simulation of flame holders in aero-engines.

2. Mathematical modeling

This article is based on the AECSC (Aero Engine Combustor Simulation Code) program, which is a simulation software developed by the Numerical Simulation Center of the School of Energy and Power Engineering at Beihang University which is based on the LES-TPDF method. It is suitable for simulating the turbulence environment in the combustion chamber of aero-engines and has been validated in numerical simulations of various configurations of combustion chambers [32].

The LES-TPDF method is a powerful modeling approach to study non-constant turbulent combustion by solving directly for large eddies on the flow, solving the PDF equations for components and enthalpy on the reaction, and solving the equation closure problem at finite reaction rates [31].

In this paper, LES is box filtered with a grid scale filter width. The Favre average control equations for mass, momentum, enthalpy, and components are expressed as:

$$\frac{\partial \bar{\rho}}{\partial t} + \partial \frac{\bar{\rho} \tilde{u}_i}{\partial x_i} = \bar{S}_{mass} \quad (1)$$

$$\partial \frac{\bar{\rho} \tilde{u}_i}{\partial t} + \partial \frac{\bar{\rho} \tilde{u}_i \tilde{u}_j}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \bar{\sigma}_{ij} - \frac{\partial}{\partial x_j} \tau_{ij}^{sgs} + \bar{S}_{mom,i} \quad (2)$$

$$\partial \frac{\bar{\rho} \tilde{\phi}_\alpha}{\partial t} + \partial \frac{\bar{\rho} \tilde{u}_j \tilde{\phi}_\alpha}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\bar{\rho} D \frac{\partial \tilde{\phi}_\alpha}{\partial x_j} \right] + \overline{\rho \dot{\omega}_\alpha(\phi)} - \frac{\partial J_\alpha^{sgs}}{\partial x_j} + \bar{S}_\alpha \quad (3)$$

$\alpha = 1, \dots, N_s + 1$, representing the component mass fractions and total enthalpy, and the set of scalar drops $\phi = \{\phi_1, \phi_2, \dots, \phi_{N_s}\}$. The dynamic Smagorinsky model solves the subgrid stress tensor [23].

In order to close the transport term J_α^{sgs} due to subgrid pulsations in Eq. (3) with the chemical reaction rate term $\overline{\rho \dot{\omega}_\alpha(\phi)}$, the literature [31] derived the equation for the probability density function $\tilde{P}_{sgs}(\psi)$ of the scalar:

$$\bar{\rho} \frac{\partial \tilde{P}_{sgs}(\psi)}{\partial t} + \bar{\rho} \frac{\partial \tilde{P}_{sgs}(\psi)}{\partial \psi_\alpha} = \mathcal{L} - \sum_{\alpha=1}^{N_s} \frac{\partial}{\partial \psi_\alpha} \left[\bar{\rho} \dot{\omega}_\alpha(\psi) \tilde{P}_{sgs}(\psi) \right] - \frac{\partial}{\partial x_j} \left[\left(\frac{\mu}{\sigma} + \frac{\mu_{sgs}}{\sigma_{sgs}} \right) \frac{\partial \tilde{P}_{sgs}(\psi)}{\partial x_j} \right] - \frac{\bar{\rho}}{2\tau_{sgs}} \sum_{\alpha=1}^{N_s} \frac{\partial}{\partial \psi_\alpha} \left[(\psi_\alpha - \tilde{\phi}_\alpha) \tilde{P}_{sgs}(\psi) \right] + \bar{S}_\alpha \quad (4)$$

The coupling between the gas-liquid phases is achieved through the source term in the equation. The liquid-phase fuel droplets are considered discrete spherical terms in the Lagrangian coordinate system, and their properties are only related to the radius, droplet number, velocity, and temperature. The filtered droplet probability density function $\bar{P}$ control equation was derived and obtained as:

$$\frac{\partial \bar{P}}{\partial t} + \frac{\partial a_j \bar{P}}{\partial V_j} + \partial \frac{\bar{N} \bar{P}}{\partial N} + \partial \frac{\bar{R} \bar{P}}{\partial R} + \partial \frac{\bar{\Theta} \bar{P}}{\partial \Theta} = 0 \quad (5)$$

The last three terms of Eq. (5) represent the changes in particle number, radius, and temperature due to fragmentation, evaporation, and convective heat exchange with the surrounding airflow, respectively, thus enabling the description of the liquid phase.

For LES, Eq. (5) is not closed due to subgrid fluctuations. However, Bini [33] found that the temperature change rate is not as important as the mass evaporation rate in studying the evaporation process of single droplets. In fact, the rate of temperature change is only important in the combustion state. Therefore, the current work ignores the influence of subgrid velocity fluctuations on the rate of droplet temperature change, and only considers the impact on the fragmentation and motion processes.

The evaporation model in this article is the A-S model proposed by Abramzon and Sirignano in 1989 [34], and the fragmentation model is the random fragmentation model proposed by Jones and Lettier, which introduces Monte Carlo trajectory integration and describes the changes in particle diameter and number caused by particle fragmentation through statistical methods. The detailed derivation process can be found in reference [35]. In terms of motion, Gardiner [36] established an equivalent stochastic ordinary differential equation system in 1983 to describe the trajectory of random particles in phase space. The acceleration of each particle is equivalent to the mass force of Maxey and Riley formula [37] plus a random Markov model, which is used to represent the impact of unresolved subgrid velocity fluctuations on a_j , thereby changing the particle's motion process and distribution:

$$dv_p = \frac{\tilde{v} - v_p}{\tau_p} + \sqrt{C_0 \frac{k_{sgs}}{\tau_t}} dW \quad (6)$$

Where, v_p is the velocity of the p th particle, $\tilde{v}$ is the gas-phase velocity at the position of the filtered particle, k_{sgs} is the gas-phase subgrid turbulent energy, C_0 is the model constant, dW represents the increment of the random Wiener process, τ_t is the subgrid time scale, which determines the interaction rate between particles and turbulent motion, defined as:

$$\tau_t = \tau_p \left(\frac{k_{sgs}}{\Delta / \sqrt{k_{sgs}}} \right)^{0.6} \quad (7)$$

Where, τ_p is the relaxation time of a particle, defined as :

$$\tau_p^{-1} = \frac{3}{8} \frac{\rho_g}{\rho_l} C_D \frac{\tilde{v} - v_p}{r_p} \quad (8)$$

In Eq. (8), r_p is the radius of the particle, C_D is the drag coefficient of the particle, with the following values [38]:

$$C_D = \begin{cases} \frac{24}{Re} \left(1 + \frac{Re^{\frac{2}{3}}}{6} \right), & 0 < Re < 1000 \\ 0.44, & Re \geq 1000 \end{cases} \quad (9)$$

In summary, the Re of a particle is determined by the relative velocity between r_p and the gas-liquid phase, thus fully describing the motion process of the particle in the Lagrangian coordinate system.

The time and space discretization in the program are carried out using the finite volume method and the Crank Nicolson scheme, with a maximum time step limit of no $> 1e-5$ s.

3. Simulation setup for two-phase cases

In this paper, a gas-phase bluff body model is selected to study the blowoff law because the chemical reaction mechanisms significantly influence the detailed model blowoff simulation. A two-phase V-gutter flame holder model in an actual combustor is selected to study the influence of different chemical reaction mechanisms on the flame stability of the bluff body.

The two-phase bluff body conditions simulated in this paper are derived from the V-gutter flame holder on the CWTF (Combustion Wind Tunnel Facility) experimental platform of Cincinnati University [39], as shown in Fig. 1. The CWTF bench is rich in data, and Bush and Gutmark used PIV to measure the velocity fields of cold and hot states [40] and the blowout limits of different oil-gas ratios [41]. The test section in red is the focus of the simulation. In front of the test section, kerosene and air are pre-burned to generate high-temperature gas. At 254 mm in front of the trailing edge, a spray bar for spray in the relative direction is set. The total air and fuel mass flow rates determine the equivalent ratio. In order to provide accurate engineering environment simulation, two working conditions in literature [40] are selected, and detailed data are shown in Table 1.

A computational domain with a size of $0.813\text{m} \times 0.152\text{m} \times 0.076\text{m}$ is established based on the test segment, and the geometric structure is shown in Fig. 2. Only the shape of the bluff body changes in two working conditions, as shown in Fig. 3.

ICEM-CFD software [22] was used to draw a grid with 1.42 million

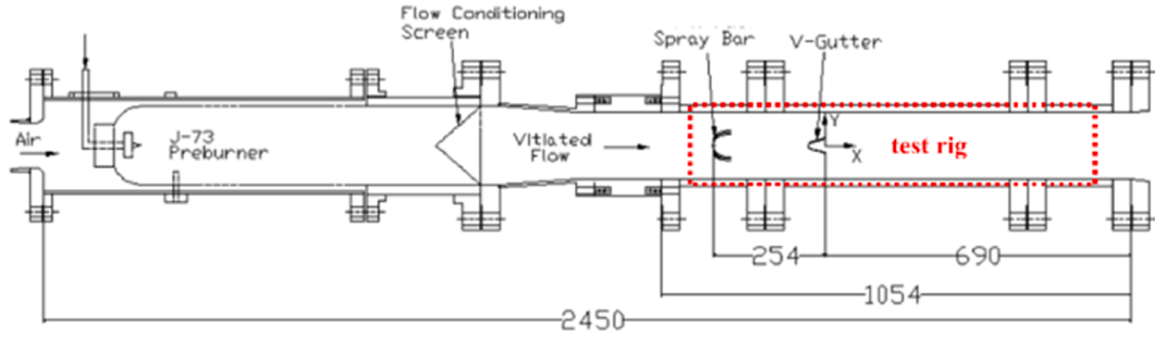


Fig. 1. Schematic of CWTF.

Table 1

Simulation condition.

Ma	Pressure /Pa	Temperature /K	Equivalence Ratio	D /mm
0.1	101,325	644	0.41	38.1
0.2	101,325	672	0.47	31.8

nodes and 36 structural blocks, and the grid number was increased on the bluff body surface, fuel injection area, and shear layer. The minimum mesh size was 0.25 mm, and y^+ was greater than 3, as shown in Fig. 4.

In order to be the same as the experiment, multiple injection points on the spray rod are retained, conical hollow sprays with a 10° Angle are used, and the fuel temperature is 301 K. Since the composition of the incoming flow in the test section is not given in the experiment, it is considered that all the kerosene is burned in the precombustor to generate CO_2 and H_2O , and the air is heated up to the average temperature of the test section, so the material composition and fuel injection amount of the test section is calculated. Atomized particle size satisfies Rosin-Rammler distribution:

$$R = 1 - \exp\left(-\left(\frac{d}{\bar{d}}\right)^n\right) \quad (6a)$$

In the equation, n is the droplet size distribution index, and $\bar{d}$ is the characteristic droplet diameter. According to the current experimental conditions [42], the spray conditions in Table 3 are obtained, and SMD accords with the distribution of measured values [43]. Detailed boundary condition Settings are shown in Table 2:

The chemical reaction mechanism requires the selection of different types and scales of reaction mechanisms suitable for the experimental Mach number range, fuel, and temperature, in order to study the simulation of the near extinction state of the blunt body stabilizer in the afterburner with different types and complexities of mechanisms. Two types of general contracting mechanisms and three types of ARC mechanisms were selected, among which JL4 and WD2 were proposed in the 1980s and have undergone many computational verifications. With the development of computing power and analytical chemistry, skeleton mechanisms have begun to be applied to numerical simulations of combustion, which has also driven research on more complex skeleton mechanisms. The Kundu mechanism, Ske61, and Z77 mechanism

have been applied in ideal reactors and turbulent combustion, thus their applicability has been verified. However, Lefebvre [49] believed that the kerosene type had little impact on combustion because most of the fuel would crack into similar low-carbon intermediates before entering the reaction zone.

The chemical reaction of the WD2 mechanism and JL4 mechanism is R_1 - R_2 and R_3 - R_6 . Franzelli [50] pointed out that the number of components determines the adiabatic flame temperature and directly influences the flow field temperature. Therefore, in Fig. 5, the chemical equilibrium and phase equilibrium models of Chemkin-Pro [37] were used to calculate the adiabatic flame temperature. In general, the five mechanisms are similar in temperature. The Kundu and frame mechanisms are both ARC mechanisms, which contain many reaction steps. The ignition delay time and laminar flame speed of the three mechanisms in Fig. 6 is consistent with the experiment.

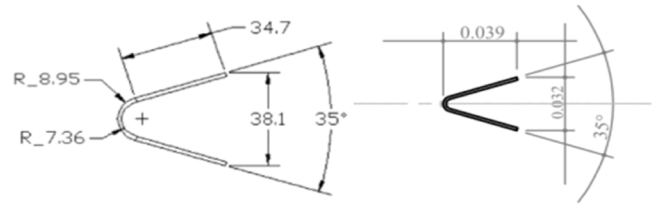
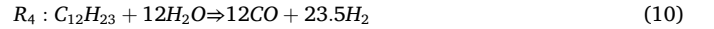
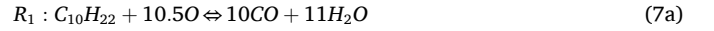


Fig. 3. Structure diagram of V-gutters (left: M0.1 condition, right: M0.2 condition).

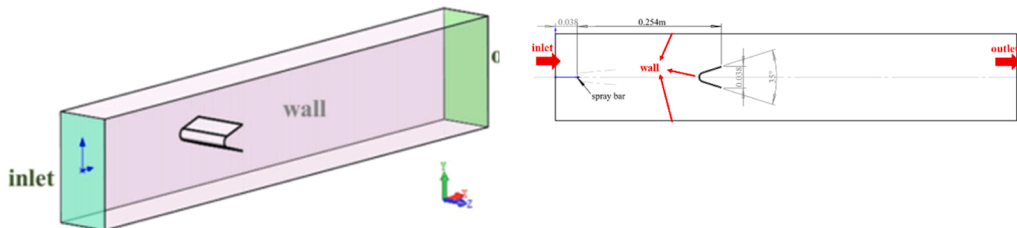


Fig. 2. Schematic of Computational model.

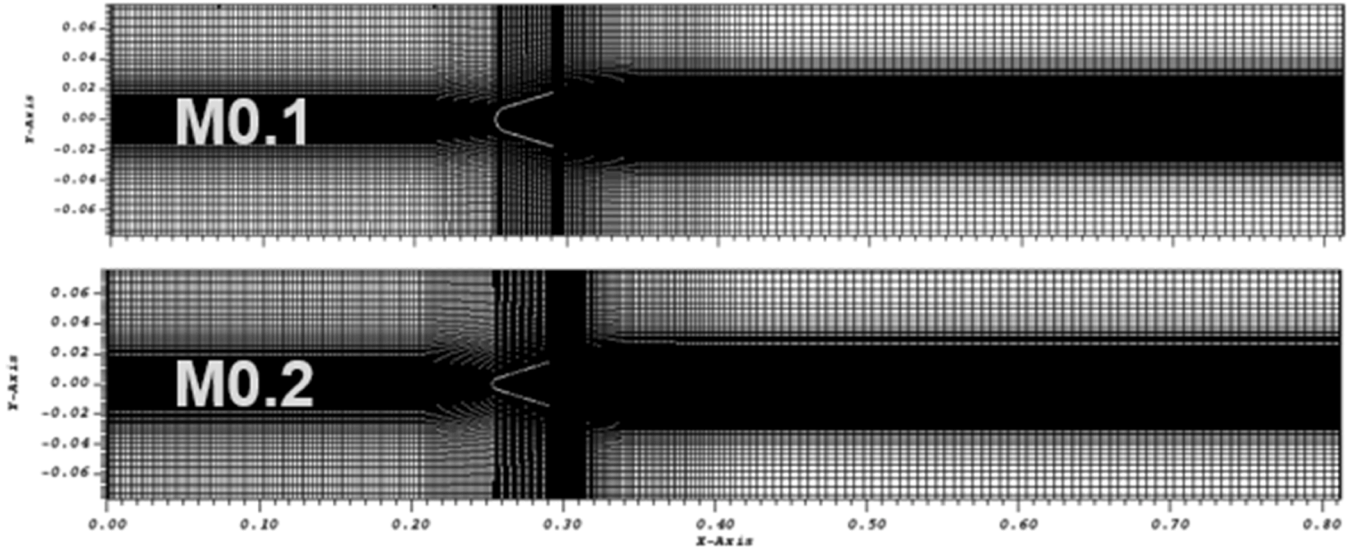


Fig. 4. Schematic of grid diagram (above: M0.1 condition, below: M0.2 condition).

Table 2

Boundary conditions.

Boundary	M0.1Setting	M0.2Setting
Inlet	Volume fraction: O ₂ 18.35 %, N ₂ 78.36 %, CO ₂ 1.68 %, H ₂ O 1.61 % Velocity: 53.3m/s	Volume fraction: O ₂ 17.80 %, N ₂ 78.54 %, CO ₂ 1.87 %, H ₂ O 1.79 % Velocity: 103.6m/s
Outlet	Pressure outlet	Pressure outlet
Wall	Adiabatic solid wall	Adiabatic solid wall
Spray	$n = 10$, $SMD=40\mu m$	$n = 5$, $SMD=25\mu m$

Table 3

Kerosene reaction mechanism.

Mechanism type	name	expression	source
Overall	WD2	n-C ₁₀ H ₂₂	The 6-component 2-step reaction mechanism of Westbrook and Dryer [44]
	JL4 _{kero}	C ₁₂ H ₂₃	Jones and Lindstedt [45] 's 7-component 4-step mechanism
ARC	Kundu	C ₁₂ H ₂₃	The 16-component 23-step reaction mechanism of Kundu [46]
	Ske61	n-C ₁₂ H ₂₆	Mechanism of 31-component 61-step reaction by Li Xiangyuan [47], Sichuan University
	Z77	C ₁₂ H ₂₃	The 30-component, 77-step reaction mechanism of Zettervall [48]

4. The analysis of Ma = 0.1 working condition

In this chapter, the steady-state combustion conditions of actual V-gutter flame holder with Ma=0.1 were simulated, and the data were compared with the experimental results. The simulated time-average results are obtained by counting the results of twice the flow characteristic time after the flow field is fully developed.

4.1. The analysis of flow characteristics

4.1.1. the comparison of TKE

Since only PIV was used to measure the velocity of the flow field in the experiment, the velocity data was selected for testing. In Fig. 7, the experimental velocity vector diagram, the simulated instantaneous velocity absolute value (unit: m/s), and the flow diagram at the back of the

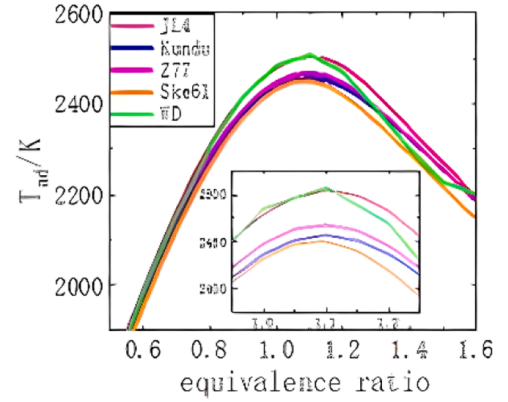


Fig. 5. Comparison of adiabatic flame temperature with five mechanisms.

bluff body are drawn. The arrows represent the direction of the incoming flow. Except for the Z77 mechanism, the simulation results of other mechanisms are similar to the experimental results after the bluff body, which can capture small vortices in the shear layer and ensure the stability of the flame. The effectiveness of the method in calculating complex turbulent combustion is proved. However, the Z77 mechanism tends to blow off, and the flame cannot be stabilized, so the vortex structure bending toward the center appears. Only four stable combustion mechanisms were compared in the following average field.

Besides, the turbulent kinetic energy (TKE), which reflected the strength of vertices and the level of the mixed action. The definition of TKE is as showed in Fig. 8. Besides, on account of the Z77 mechanism couldn't kept combustion stably, only the other four reaction mechanisms will be discussed in the following.

$$TKE = \frac{3}{4} (\overline{u^2} + \overline{v^2}) \quad (13)$$

The outline of the TKE reflects the path of the vortices shedding from the edge of the V-gutter flame holder, and the four mechanisms are symmetrically distributed in the gradually widening turbulent kinetic energy high value region with the performing the best.

Fig. 9 compares the TKE of simulation and experiment at Y/D = 0.5 in the shear layer. It can be got that the TKE of JL4_{kero} mechanism is higher than the kind of ARC mechanisms, which shows the agreement of the simulation and the experiment.

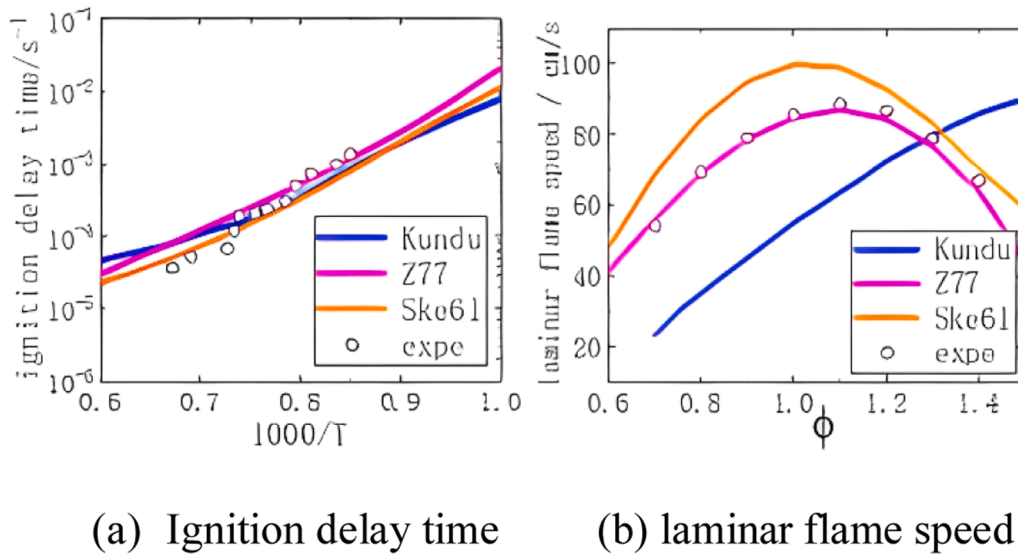


Fig. 6. Ignition delay time and laminar flame speed of ARC chemical mechanisms.

4.1.2. the comparison of the length of the reflow zone

The reflux zone reflects the effect of bluff body flame stabilization, and its length is an important parameter characterizing combustion stability. In this section, the shape of the reflux zone is determined by the zero value of the axial time average velocity. Fig. 10 compares the experimental and simulated time-averaged axial velocity cloud maps.

It can be seen that the phenomenon of reflux elongation can be got by all mechanisms although the length of the reflux zone they got are different.

Positions of axial $X/D = 0.25, 0.8, 1.3$, and 4.0 were intercepted. Fig. 11 shows the transverse distribution of time-mean axial velocity U_{mean} . The first three sections are in the reflux region, and the last is in the wake region. All the mechanisms in the reflux area are consistent with the experiment. In the wake region, the relative errors of JL4 and frame mechanism are 15.39% and 19.81%, within 20%. The method is proved to be effective in calculating complex turbulent combustion.

According to the analysis of the error, on the one hand, the high Re which reached to 33,800 made the occurrence of the vortex street, which make the flame beat back and forth. That's why the error occurred; On the other hand, different mechanisms have different reaction dynamics progresses which lead to the difference of combustion temperatures and the distribution of components, makes the influence to the flow. Besides, although ARC reaction mechanisms which have more detailed steps performed better in the ignition, propagation, and blowoff characteristics, of the 0-dimensional or 1-dimensional flame, there appear blowoff that should not appear in CFD. In addition, even the other four mechanisms of stable combustion have significant differences in the length of the reflux zone.

4.2. The analysis of the result of combustion characteristic

4.2.1. Temperature and the heat release rate

Although the time averaged temperature and composition were not measured in the experiment under M0.1 operating conditions, the composition reflects the chemical kinetics process in the current solution of incompressible flow fields. Temperature directly determines the density value through the ideal gas state equation, which affects the pressure velocity and is an important parameter in the analysis of hot flow fields. Therefore, Fig. 12 presents the instantaneous temperature and heat release rate distribution after the blunt body to indicate the flame morphology and reaction zone of different mechanisms. The black solid line in the figure represents the chemically appropriate location.

As we can see in Fig. 12, due to the mixing effect of the shear layer in

the near field of the bluff body, the reaction area with high heat release rate is stable in a band shape at the root of the bluff body. As the flow oscillates downstream and the components mix continuously, the flame expands to a chemically appropriate position, forming a wide high-temperature distribution in the wake region similar to vortex street flow, indicating strong instability of the blunt body flame. Reference [51] compared the structures of bluff body flames at different reactant temperatures and pointed out that the reason for the formation of the recirculation zone of the blunt body is that the vortex strength is weakened through expansion after combustion heat release, which promotes the stability of the flame. Therefore, the simulation of diffusion flames in the near field with different mechanisms is the key to determining the reflux characteristics, and the simulation also found that the combustion temperatures of the five mechanisms were different, with the overall package mechanism having a higher temperature of approximately 2300 K.

Fig. 13 shows the instantaneous HRR mixed fraction scatter plots of two types of mechanisms.

In summary, the flame temperature simulated by the ARC mechanism in the CFD flow field is mainly determined by the temperature at the extinction limit, rather than the theoretical adiabatic flame temperature. In addition to temperature values, it was found in the simulation that the high temperature distribution of the WD2 mechanism is particularly different from that of the Kundu mechanism, with high-temperature air masses filling the interior of the V-shaped bluff body. Therefore, it is speculated that the mechanism of kerosene varies due to different chemical reaction processes, resulting in differences in heat release and component distribution, thereby affecting the flow of high-temperature air masses and ultimately changing the morphology of the reflux zone, resulting in differences in the flow field in the previous section. Therefore, further discussions will be conducted on the composition of the flow field to deepen the influence mechanism of the mechanism.

4.2.2. Analysis of the structure of the flame and species

In the current TPDF combustion model, the biggest advantage is the direct solution of component transport and reaction rate, especially the latter, which can intuitively reflect the reaction situation of key substances in turbulent flow fields. Due to the close correlation between the flame structure and the distribution of combustion intermediates, Fig. 14 depicts the multi-component distribution of different mechanisms on the blunt body side, including kerosene, oxygen, and key components.

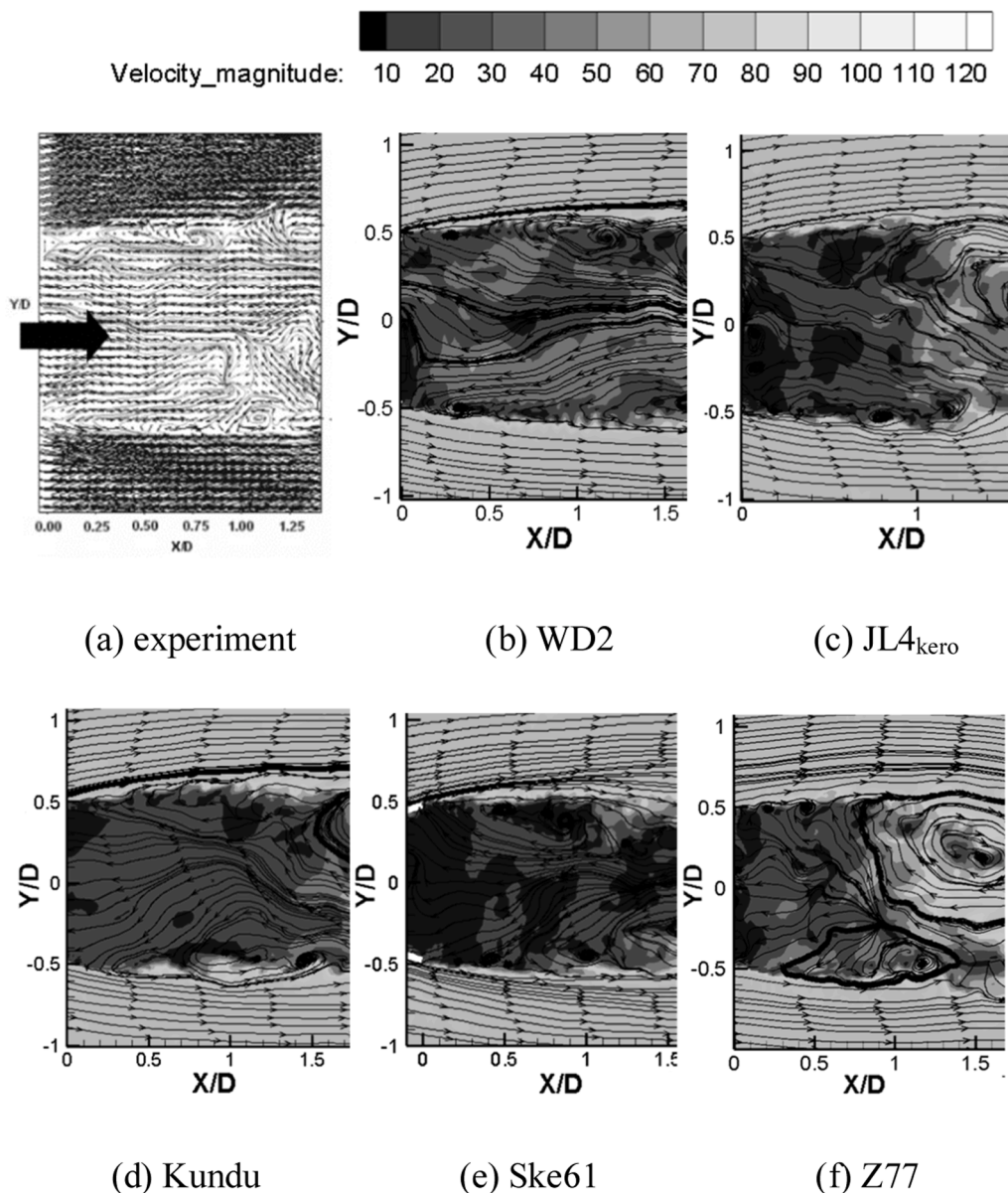


Fig. 7. Instantaneous velocity and streamline distribution of bluff body under M0.1 condition.

Among the many intermediate products of kerosene combustion, OH radicals and CH_2O content are two important substances: OH can characterize the reaction zone of the flame, and CH_2O is used to characterize the preheating zone of the flame. However, the general contracting mechanism often omits free radicals and intermediate products with lower content, retaining only the most critical substances. For example, the WD2 mechanism intermediate products selected in this article only have CO, while the JL4_{kero} mechanism intermediate products only have CO and H_2 . Therefore, this type of mechanism cannot locate the reaction or preheating zone based on the distribution of OH radicals and CH_2O . In Fig. 14, the simplified component is selected to reflect its flame structure, and the reactants, intermediate products Group display of end products.

In the preheating stage, the pyrolysis process of ARC type mechanism and general package type mechanism kerosene is different as shown in Figure(c), (d), (e). The former considers the process of fuel splitting into small molecule hydrocarbon products, where large molecules of fuel first break into multiple C2 or C3 hydrocarbons and generate a large number of free radicals. These substances further react to obtain higher activity low carbon number small molecule intermediate products,

where CH_2O is a representative substance, indicating that the fuel has been fully pyrolysis and is about to enter the subsequent exothermic reaction. Among the three ARC mechanisms, the Kundu mechanism has the simplest pyrolysis process, with $\text{C}_{12}\text{H}_{23}$ generating CH through a one-step reaction. Therefore, in Fig. 14(c), CH is used instead of CH_2O . The Kundu and Ske61 mechanisms that can stabilize combustion distribute a large amount of CH or CH_2O in the reaction zone of the shear layer, indicating that kerosene undergoes sufficient pyrolysis after preheating; As shown in Figure (e), the CH_2O concentration in the preheating zone of the Z77 mechanism is too low, and the kerosene is not fully pyrolyzed. The generation of CO and H_2 depends on these small molecule products, so it is not possible to generate sufficient CO and H_2 . The pyrolysis of the general package mechanism is different from the ARC mechanism, as it omits the process of fuel cracking into small molecule high activity products and directly obtains the key intermediate products CO and H_2 , thus the pyrolysis process is faster. Especially taking the JL4_{kero} mechanism in Fig. 14(b) as an example, the concentration of CO or H_2 is high.

After the pyrolysis of kerosene in the preheating stage, combustion enters the stage of oxidation and heat release. The three ARC

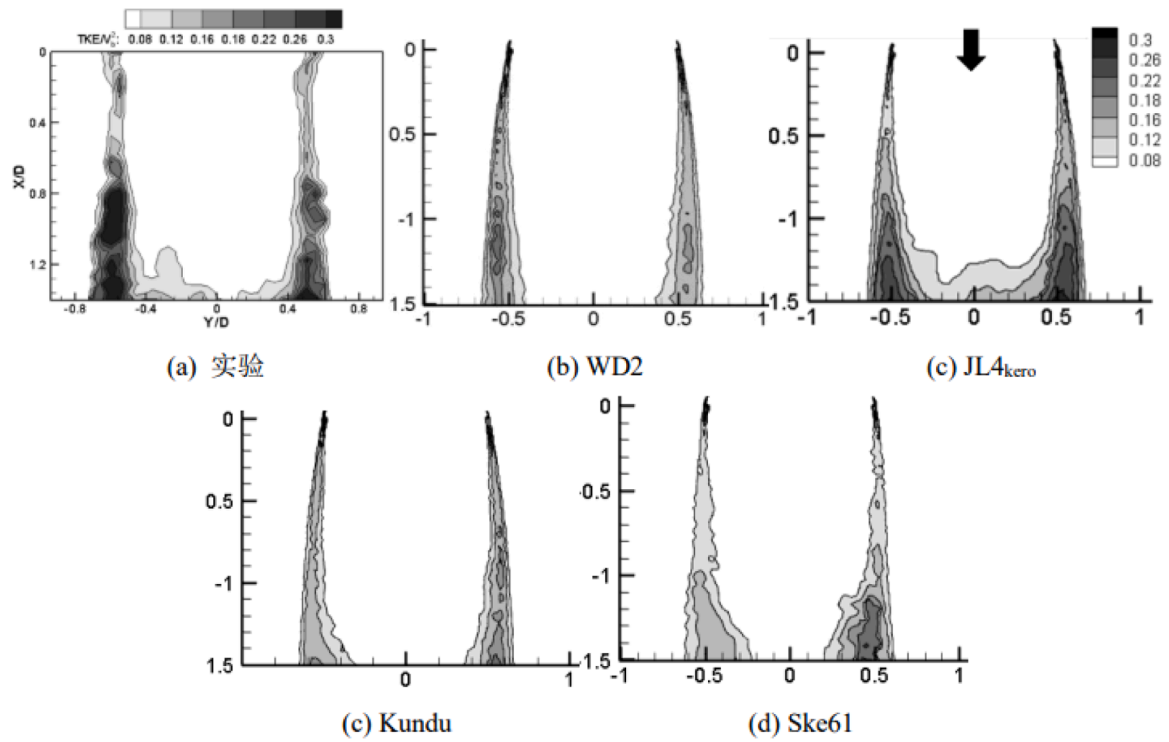


Fig. 8. The distribution of TKE behind the bluff body (M0.1).

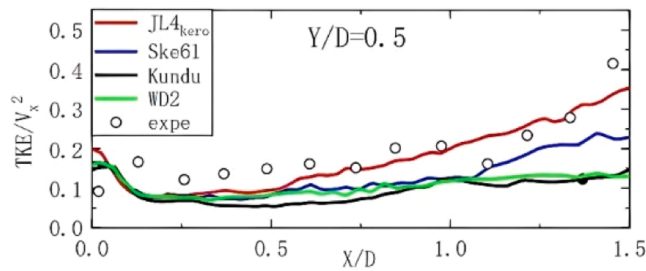


Fig. 9. The distribution of TKE in shear layer (M0.1).

mechanisms all contain OH molecules. During this process, OH, as the most critical free radical, participates in the chain transfer and termination of the reaction, providing a large number of highly active substances for the combustion system. On the one hand, it oxidizes the low carbon number small molecule intermediate products generated in the preheating stage to generate CO or H₂, and on the other hand, it directly participates in the bond exothermic reactions R7 and R8 in Section 4.2.1 to generate CO₂ or H₂O. The Kundu and Ske61 mechanisms also exhibit high concentrations of OH in the reaction zone, indicating active reaction; The low OH concentration and weak exothermic reaction in the Z77 mechanism are the reasons for the low instantaneous heat release rate and temperature in the previous section. The general contracting WD2 and JL4_{kero} mechanisms also achieve higher heat release rates and

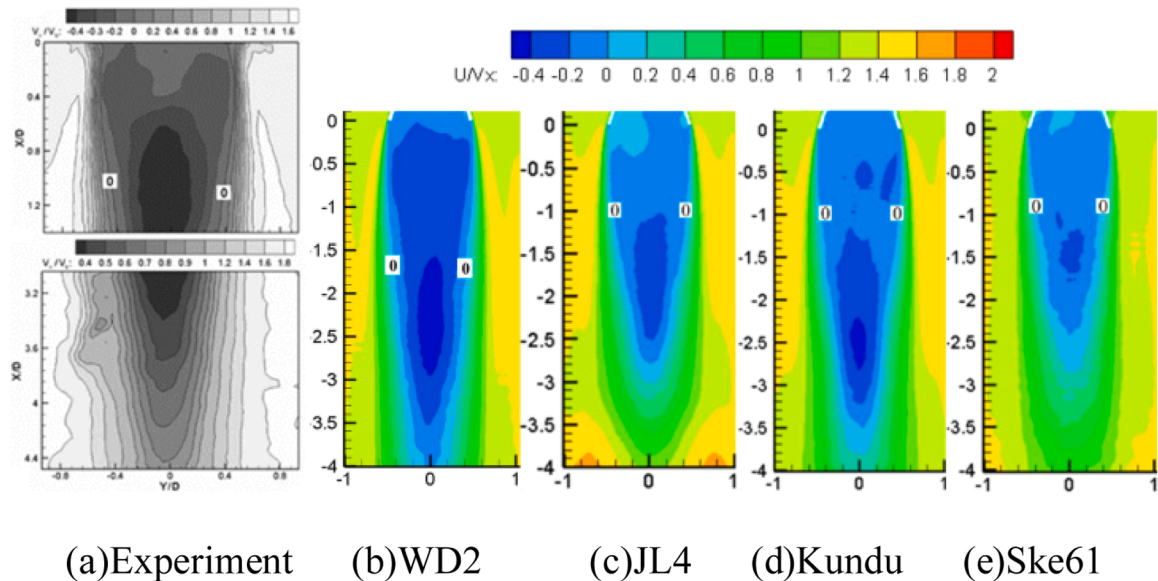


Fig. 10. The distribution of time-average velocity behind the bluff body (M0.1).

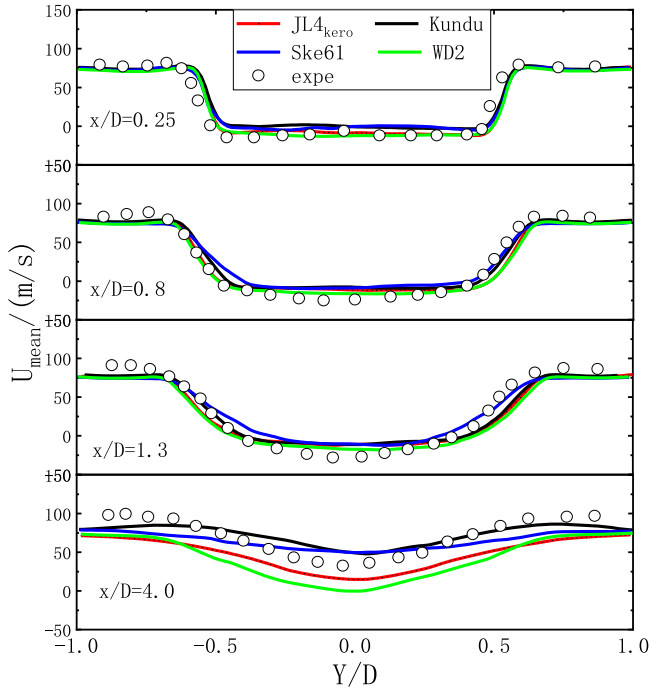


Fig. 11. Flow velocity comparison diagram.

temperatures in the flow field due to their simple exothermic reaction and high heat release.

Glassman [52] pointed out that both general contracting and ARC type reaction models involve the oxidation of C and H in high-temperature hydrocarbon combustion, where CO and H₂ are important intermediate products, and their oxidation is synchronized with the high exothermic process of the system. Therefore, the balance between CO and CO₂, H₂ and H₂O is the key to the reaction. Therefore, the balance between CO and CO₂, H₂ and H₂O is the key to the reaction. Therefore, Fig. 15 shows the time-average mass fraction and generation rate of CO, H₂ and product CO₂, H₂O located at X/D = 1.0 after the bluff body.

Y/D is located in the rich oil zone on the outer side of the bluff body, the shear layer, and the reflux zone at the center, from large to small. After passing through the reaction zone of the shear layer, the concentration of each substance increases, and CO and H₂ are subsequently

consumed, resulting in a decrease in concentration in the reflux zone. CO₂ and H₂O, as relatively stable products, still maintain high concentrations in the reflux zone. From the analysis of generation rate, all four substances mentioned above have a positive generation rate in the reaction zone, indicating that their generation amount exceeds their consumption, so they can be accumulated in the reflux zone.

In contrast, the Kundu mechanism has a high concentration of CO in the reflux zone behind the bluff body, with a large amount of CO accumulating; The overall WD2 mechanism ignores H₂ and therefore relies more on the reaction R2 of CO. The high concentration of CO₂ in the reflux zone indicates that a large amount of CO is oxidized to CO₂. The CO₂ and H₂O generation rates at the reaction sites of these two mechanisms are relatively low, while the CO₂ and H₂O generation rates are high at the overall JL4_kero and Ske61 mechanism reaction sites. The oxidation process of CO and H₂ is the strongest, and there is no accumulation of a certain component in the reflux zone.

4.2.3. The reaction characteristic of mechanisms in flow field

Reference [37] analyzed the structure of blunt body combustion flames at different reactant temperatures and pointed out that the reason for the extension of the recirculation zone of the bluff body in the hot state is due to combustion heat release, which weakens the vortex strength due to expansion and promotes flow stability. The most critical exothermic reactions of different chemical mechanisms are the reactions of CO to CO₂ and H₂ to H₂O. The JL4_kero mechanism is implemented by reactions R5 and R6, respectively. Almost all CO₂ in ARC type mechanisms is obtained through R8 reaction, but H₂ oxidation cannot be fully represented by R7. Therefore, the reaction heat of this process is calculated using the reaction enthalpy of H₂ oxidation to H₂O. To refine the heat release characteristics of each reaction, a temperature reaction heat scatter plot of the reflux zone at the center section is plotted in Fig. 16. HRR is the total heat release rate, and R_{CO_2} , R_{H_2O} represents the reaction heat of CO towards CO₂ and H₂ towards H₂O.

In terms of the heat release rate distribution of the reaction, the four mechanisms of R_{CO_2} and R_{H_2O} are all relatively high throughout the entire temperature range, confirming that they are important exothermic reactions. Compared with other mechanisms, the R_{CO_2} and R_{H_2O} of WD2 and Kundu mechanisms are lower, indicating weak oxidation of CO and H₂. Therefore, the unconsumed CO and H₂ in the reaction zone are drawn into the reflux zone, resulting in excessively high concentrations in the reflux zone. These substances slowly release heat internally, promoting flame stability, and making the reflux zone too long and simulation deviation large.

To calculate the heat release ratio of the different oxidation processes

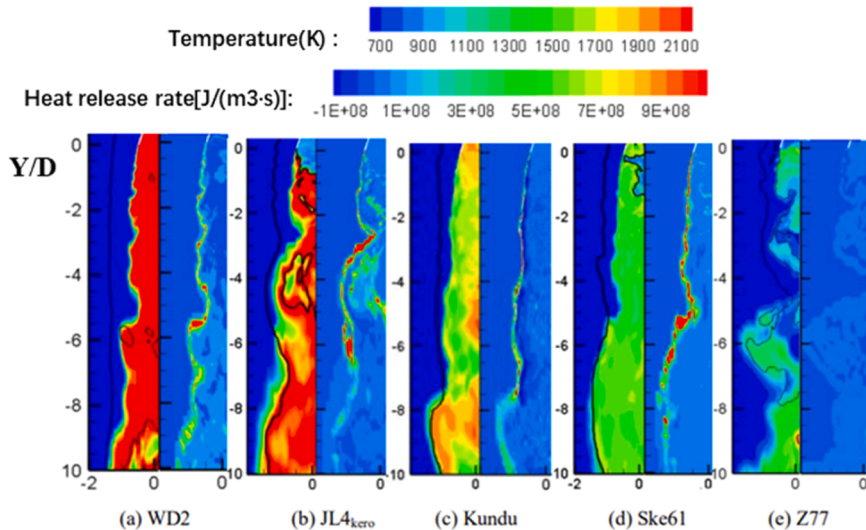


Fig. 12. The distribution of the instantaneous temperature and heat release rate diagram.

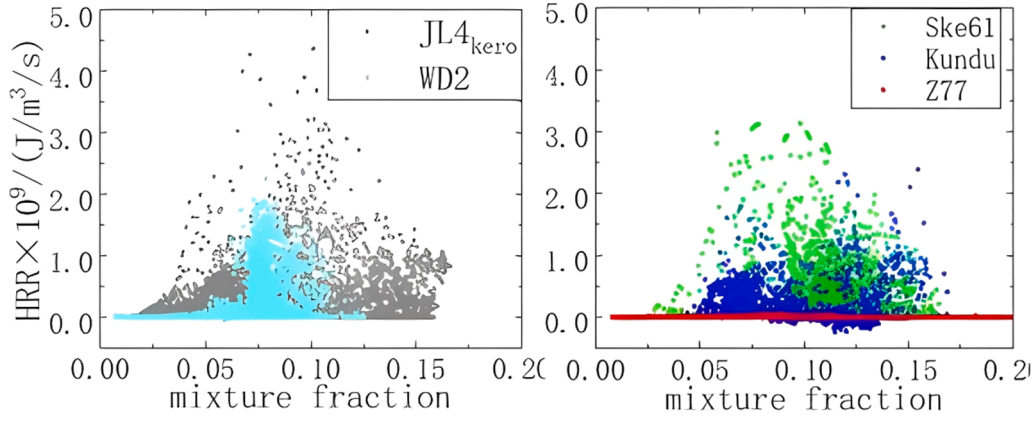


Fig. 13. Scatter plot of instantaneous heat release rate and mixing fraction under different mechanisms (M0.1).

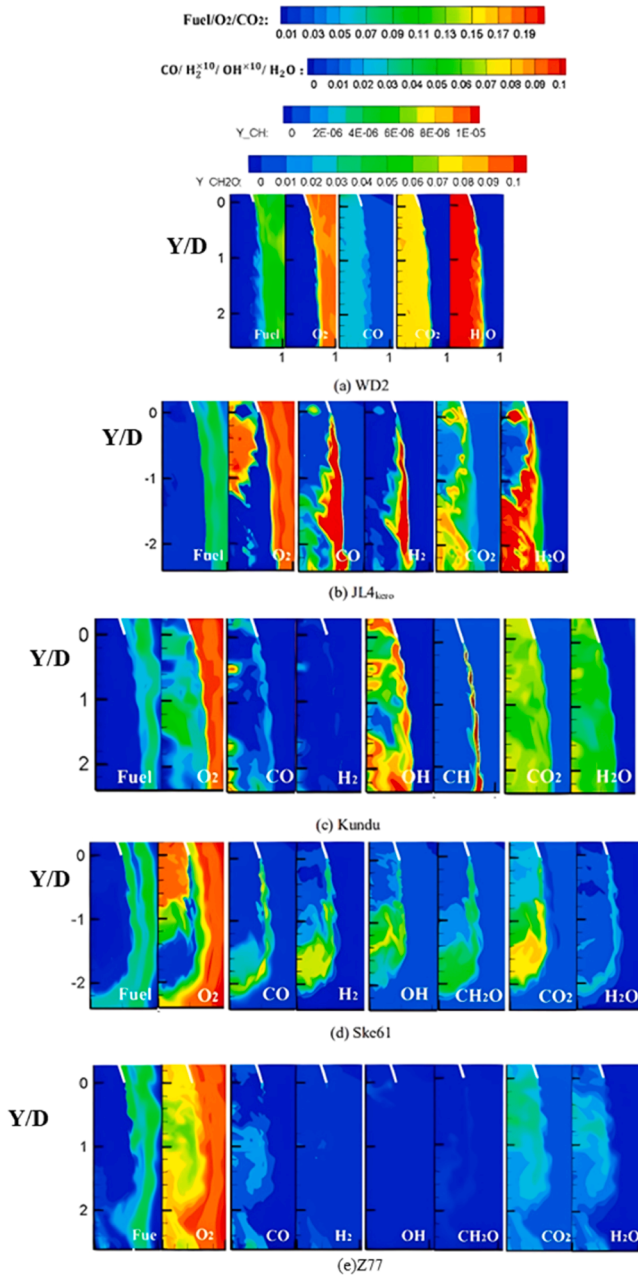


Fig. 14. Different flame structure based on different mechanisms (M0.1).

mentioned above, a high temperature gradient value (greater than $2 \times 10^5 \text{ J/m}^3/\text{s}$) was used to intercept the intense chemical reaction zone shown in Fig. 12, multiple cross-sectional positions within the reflux zone were taken to calculate the average heat release of different reactions, and the impact of each reaction was analyzed from the perspective of heat release. Fig. 17 shows the average HRR values of the WD2, JL4_{kero}, Kundu, and Ske61 mechanisms at cross section positions $X/D = 0.25, 0.8, 1.3, 1.8,$ and 2.3 (represented by pure colors of orange, green, purple, and blue), and the heat release rates of CO and H₂ oxidation reactions are represented by two oblique encryption methods of light colored thick and thin lines. Two stacking bars are defined to represent the heat release rate of this process. The definition of the heat proportion is as shown in Eq. (14):

$$\eta_{HC} = \frac{\eta_{HC} + \eta_{HC}}{HRR} \quad (14)$$

The heat release rate column chart of the reaction layer quantifies the strength of reaction heat release, and the reaction intensity of different mechanisms is directly related to chemical expressions and reaction rate coefficients. The exothermic characteristics of CO and H₂ oxidation reactions are different between the general contracting and ARC mechanisms. In the general contracting mechanism, WD2 does not consider the H₂ reaction, so only the oxidation heat release of CO is counted; The JL4_{kero} mechanism relies almost entirely on the R7 and R8 reactions due to its few components and simple reaction. At the same time, the heat release of the H₂ reaction is much stronger than that of the CO reaction. Among ARC mechanisms, Kundu mechanism's R_{CO2} and R_{H2O} are the smallest, and reaction R7 and R8 are weak; The oxidation heat release of H₂ and CO in the Ske61 mechanism is similar.

5. The analysis of Ma = 0.2 working condition

5.1. The analysis of the result of stable combustion

The simulation of M0.2 working condition has the same reaction mechanism as the previous chapter. As the Reynolds number increases, the blunt body combustion environment becomes more harsh and the flame is not easy to stabilize. Simulation shows that only the JL4_{kero} mechanism can obtain flames that are consistent with the experiment at this time, while other mechanisms exhibit extinguishing problems similar to the Z77 mechanism under M0.1 working condition.

The flow rate under the M0.2 working condition is 0.2 Ma in practical engineering applications, with a Reynolds number of up to 51,100, and also has the characteristic of poor oxygen. Compared with the M0.1 working condition, turbulence makes the flow oscillation behind the blunt body stronger. Fig. 18 shows the distribution of Z-direction vorticity ω_z for the JL4_{kero} mechanism under two working conditions, and quantitatively analyzes the vortex intensity in the near field reflux

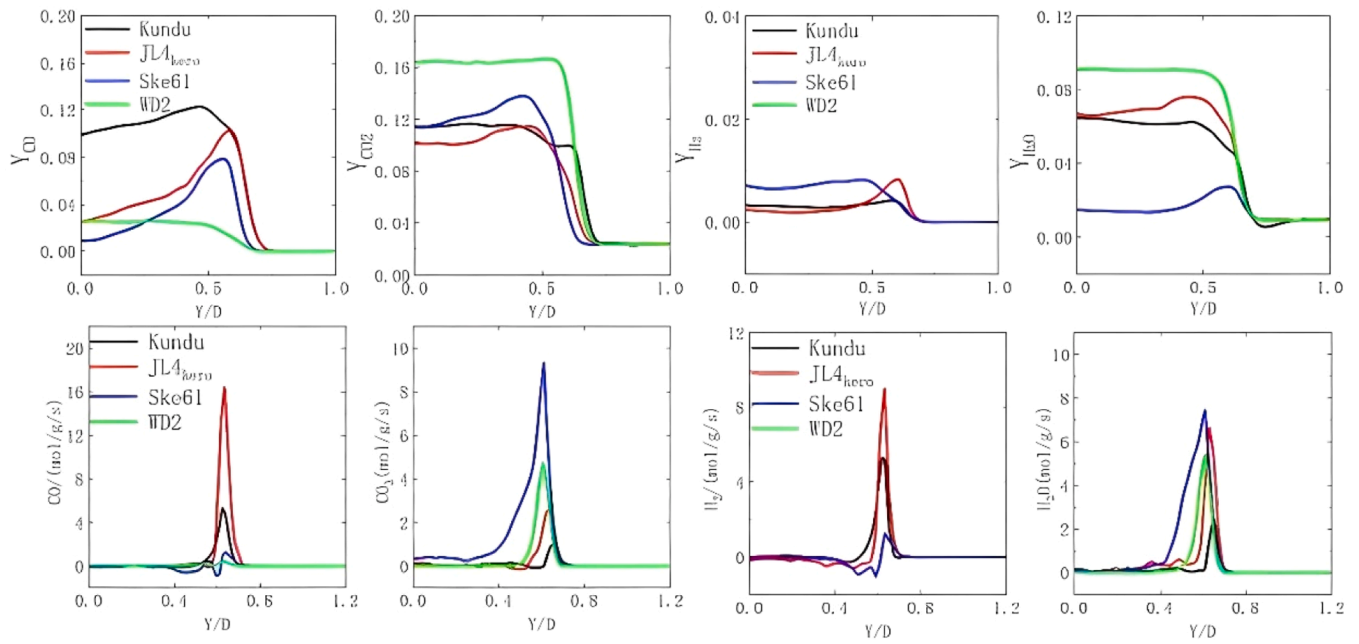


Fig. 15. Horizontal distribution map of CO, CO₂, H₂, H₂O mass fraction and generation rate (upper: mass fraction, lower: generation rate).

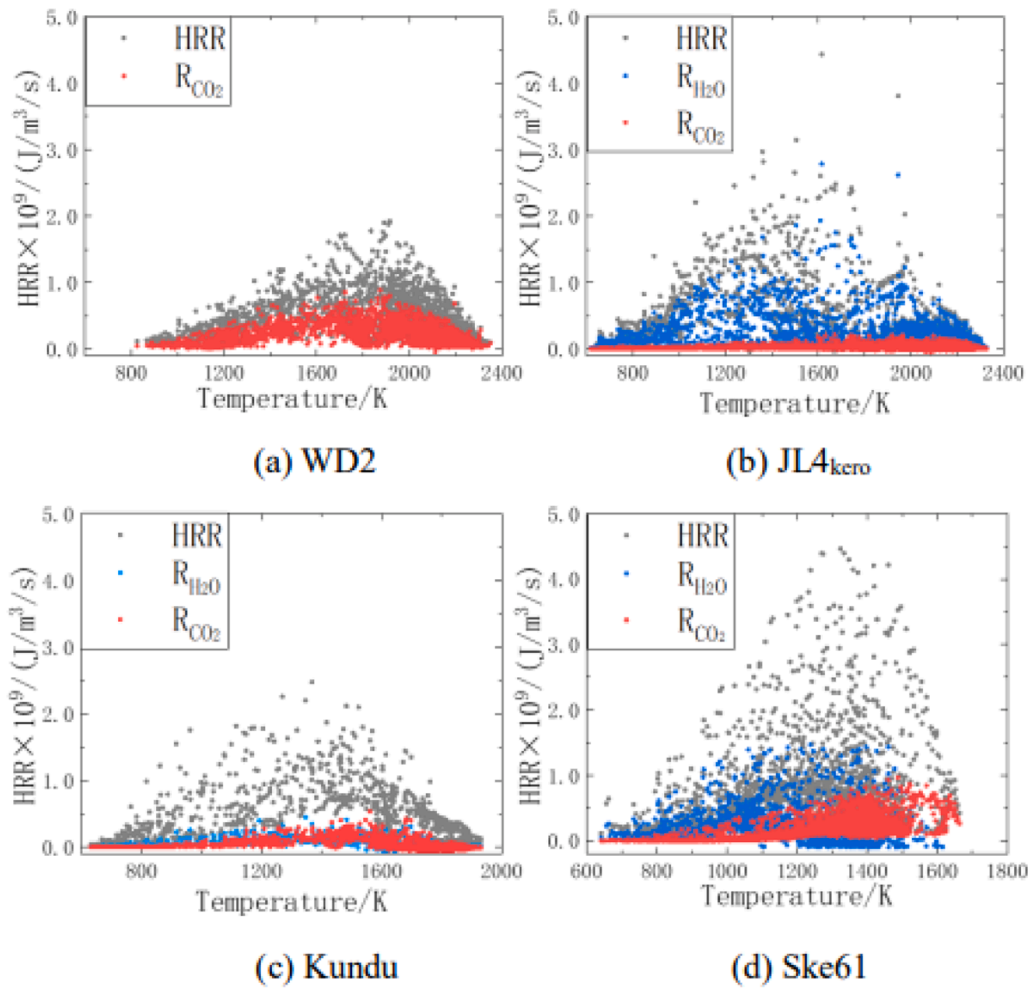


Fig. 16. Flow velocity comparison diagram.

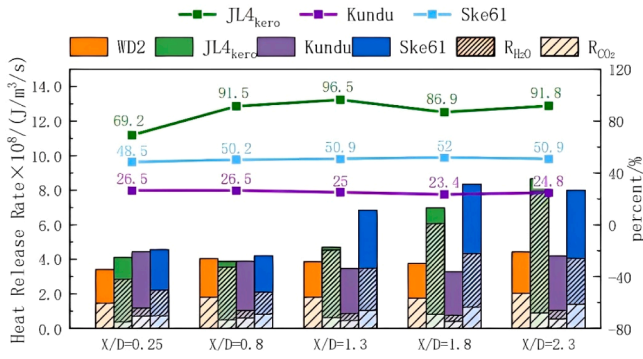


Fig. 17. Heat statistics of CO and H₂ oxidation reaction in the reaction layer (M0.1).

region of the bluff body:

The black solid line represents a high heat release rate of 5×10^8 J/(m³·s) contour line, used to identify chemical reactions. When the flow rate increases, the intense reaction area exhibits intermittent distribution (indicated by the arrow in the figure), which increases the difficulty of flame stabilization by different mechanisms. And at high speeds, the vorticity in the flow field increases, leading to the appearance of two sets of asymmetric reverse vortices behind the bluff body, which rapidly detach from the rear of the bluff body and form a vortex street. In actual simulation, only the JL4_{kero} mechanism can stabilize combustion.

The instantaneous temperature gotten by experiment and JL4_{kero} are shown in Fig. 19.

It can be seen in Fig. 18 that the distribution of instantaneous temperature got by simulation is similar to the result in experiment. The flame symmetrically distributed along the root of the bluff body, which signals it generated and stabilized in the reflux zone. And the flame started to swing when it entering the swing zone. Compared with the M0.1 working condition, the more unstable the combustion is, the intenser the vortex street will be. Furthermore, the stability of flow increased sharply due to the weaker expansion, whose reason is the temperature drop of the reflow area.

Then, the shedding of the high temperature vortex blobs caused by the increased flow instability ignited rich-oil gas. As the result, the high temperature zone whose highest temperature will achieve 2300 K will appear at far-end of the swing zone.

In Fig. 20, the horizontal time-averaged temperature is compared, and the three cross-sections along the flow direction are located in the reflux zone, the disappearance of the reflux zone, and the far-field wake zone. The simulated relative errors were 22.11 %, 11.57 %, and 13.92 %, respectively.

The first cross-sectional error source analyzed is the high peak reaction temperature of the flame at the root of the blunt body, while the second two cross-sectional errors originate from the wider flame. Overall, the lateral distribution trend of the average temperature in the JL4_{kero} mechanism is consistent with reality. Although flames with a general contracting mechanism are stronger than actual ones, a reasonable temperature distribution can still be obtained in the simulation of strong turbulent combustion fields.

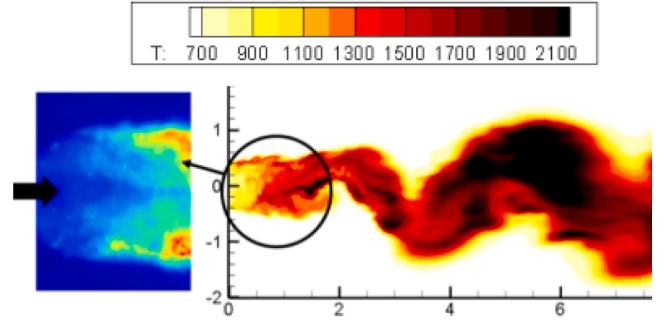


Fig. 19. The instantaneous temperature under M0.2 (left: experiment; right: simulation).

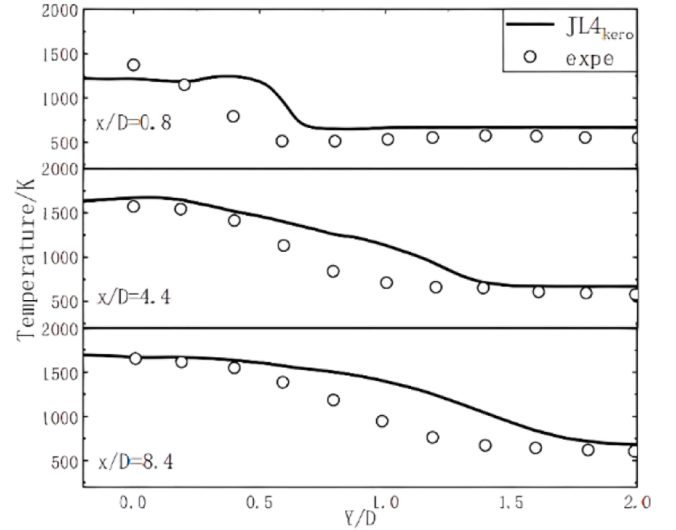
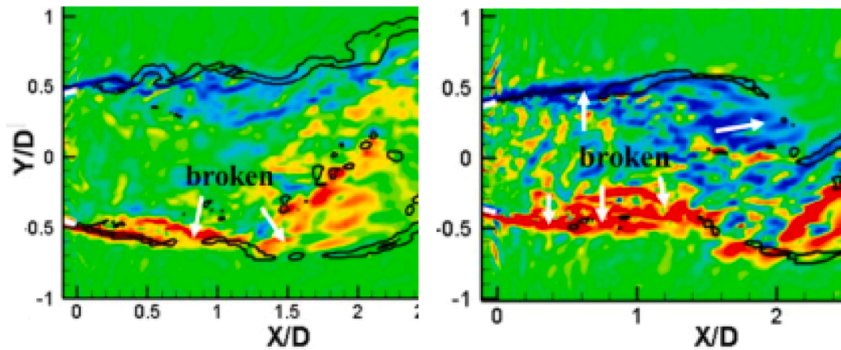


Fig. 20. The transverse distribution of time-average temperature (M0.2).



(a) M0.1

(b) M0.2

Fig. 18. Distribution of spanwise vorticity under different operating conditions.

Table 4
Comparison of combustion conditions of kerosene simplification mechanism.

condition	WD2	JL4 _{kero}	Kundu	Ske61	Z77
M0.1	✓	✓	✓	✓	×
M0.2	×	✓	×	×	×

5.2. The analysis of the reason of blowoff

Similar to the Z77 mechanism in the M0.1 condition, more mechanisms could not form a stable flame in the M0.2 condition. This phenomenon indicates that the chemical mechanism greatly influences the

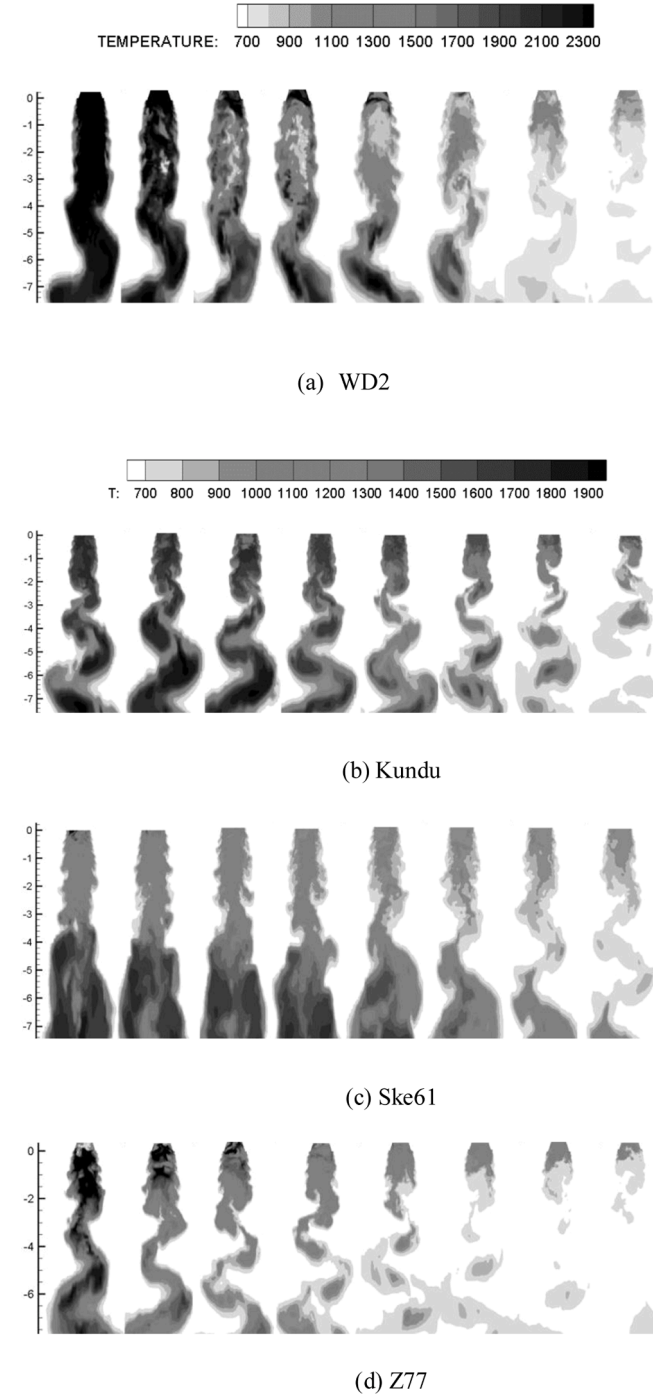


Fig. 21. Instantaneous temperature distribution of different mechanism blow-off process.

simulation of bluff body flame. The simplified mechanism of accurate ignition delay time and flame propagation velocity does not necessarily apply to high-speed bluff body combustion. In Table 4, the combustion conditions of the five mechanisms are showed. "✓" means the flame can be stationary, while "×" means it cannot. In the simulation, the steady-state combustion results of a simple mechanism, common in engineering, are used as the initial flow field for calculation. The time interval of the WD2 and Kundu mechanisms is 1.2 ms, and the time interval of the Ske61 and Z77 mechanisms is 0.4 ms.

In Fig. 21, all blowoff mechanisms first experience a temperature drop and length shortening in the reflux zone, and then the flame is disconnected from the middle. After the mechanism changes, the temperature in the near field of the bluff body drops rapidly, and the temperature of the vortex shedding from the reflux area is too low to ignite the gas in the downstream flow field. Later, the flame becomes more and more difficult to maintain stability due to the influence of vortex street flow caused by the wake swing and finally breaks off. In comparison, the blow-off speed of WD2 and Kundu is slow, while Ske61 and Z77 are fast, and Z77 is extremely fast. This is because the combustion temperature of the WD2 and Kundu mechanisms is high, and the high temperature of about 2000 K can be stable in the reflux region. Therefore, the temperature of the alternating shedding vortex is higher, and the gas behind ignited by it briefly maintains the flame of the wake.

In the flow field, the high strain rate is mainly distributed in the root of the bluff body, high in the shear layer on both sides, low in the central reflux area, and decreasing along the downstream flow. In order to further study the cause of blowoff and judge the local flame combustion state, the flow strain rate K_s under M0.1 and M0.2 conditions was analyzed, and the local combustion state was judged by K_s combined with the blowoff limit of the mechanism. The flow strain rate K_s is defined in the literature [9], where n_x and n_y are the directional components of the flame surface. Then select the combustion reaction layer based on HRR and take it as the flame surface, calculate n_x and n_y based on the flame surface morphology, and retain the high heat release area to calculate the flow strain rate K_s :

$$\kappa_s = \left| -n_x \times n_y \times \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) + (1 - n_x^2) \times \frac{\partial u}{\partial x} + (1 - n_y^2) \times \frac{\partial v}{\partial y} \right| \quad (15)$$

In numerical simulation studies, Karlovitz (Ka) number is often used to judge local blowoffs. Ka number is the ratio of K_s to K_{ext} . When $K_s < K_{ext}$, the flame can exist stably, while when $K_s > K_{ext}$, it blows off.

Fig. 22 calculates the probability density of K_s in the high-reheat release region ($HRR > 2 \times 10^8 \text{ J/(m}^3\text{/s)}$) of four mechanisms under two working conditions and makes Gauss fitting. In the figure, the solid black line represents the Gauss fitting curve of probability density, and the black dotted line is the K_{ext} obtained by experiment [53] when $\phi = 1.2$. At this time, the K_{ext} of high-carbon alkane reached the maximum, about 1250 s^{-1} . WD2, Kundu, and Ske61 mechanisms intercept the time before blowoff in M0.2 working conditions. However, in the Z77 mechanism, due to the speedy blowoff process and low heat release rate, there was almost no severe reaction zone, and K_s in the reaction layer could not be counted. In the M0.1 condition, 36.4, 50.5, 41.0, and 50.2 % of the simulation results of the WD2, JL4, Kundu, and Ske61 mechanisms are higher than the experimental value of 1250 s^{-1} . Therefore, in the stable combustion state, less than half of the high heat release reaction zone is still in the unstable combustion state, and the tensile effect caused by high-speed flow will affect the flame and make it tend to extinguish.

Fig. 23 compares the fitted K_s probability density function curve under two velocity conditions. As the velocity increases, the region with a low strain rate decreases while the region with high strain rate increases, and the K_s probability density curve becomes more gentle. Under the M0.2 condition, 57.9, 60.1, 53.7, and 53.9 % K_s of WD2, JL4, Kundu, and Ske61 are higher than the experimental value. The proportion of unstable combustion zone is higher than $M = 0.1$ condition, indicating that more areas in the flow field are unstable, and the local

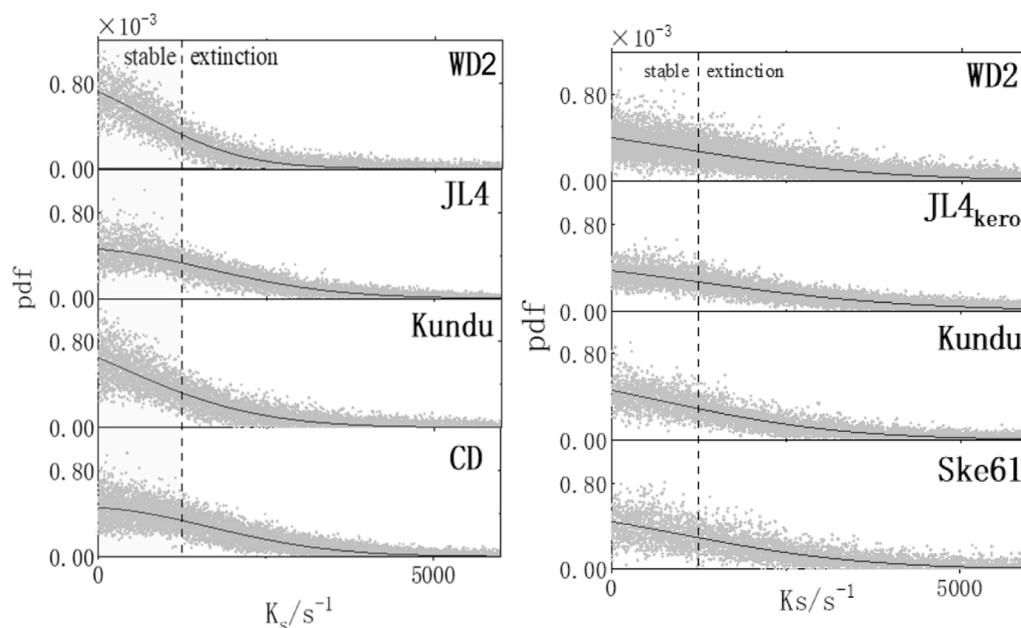


Fig. 22. Probability density distribution of K_s (left: $M = 0.1$ condition, right: $M = 0.2$ condition).

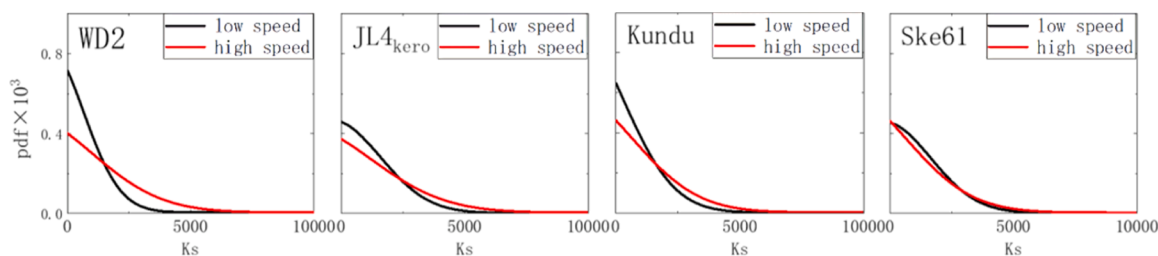


Fig. 23. Comparison of K_s probability density curves in high and low speed conditions.

combustion state tends to extinguish.

In Fig. 24, the transverse temperature, heat release rate, and molar concentration distribution of each component on the unilateral reaction layer of the bluff body are captured at the flow direction $X/D = 1.0$. Position Y shows the transverse unreacted oil-rich gas and the reaction products from large to small, and the concentration of each substance is represented in a logarithmic coordinate system.

For the ARC mechanisms, kerosene is first decomposed in the preheating stage, and then small molecular products are finally oxidized into CO , H_2O , and CO_2 and release heat. The above two processes can be separated on a spatial scale, and the temperature in the preheating zone can reach >1000 K, which is high enough to ensure the pyrolysis reaction. At the same time, the pyrolysis process provides great H and OH free radicals, and the reaction of the substance with the fuel molecules further promotes its decomposition and increases the temperature. The reaction and temperature rise promote each other and accumulate many reactants for the subsequent heat release stage.

In Fig. 24, for the ARC mechanisms of Kundu, Ske61, and Z77 under $M=0.1$ condition, with the decrease of fuel concentration, free radicals such as OH and H and pyrolysis products such as C_3H_6 and C_2H_3 increase significantly, resulting in primary rise in temperature. Later, as the pyrolysis products are oxidized, the concentration of final products like CO_2 and H_2O increases, and the heat release rate reaches the highest value. Although the Z77 mechanism of unstable combustion also has the above phenomenon, the content of free radicals (such as OH) and pyrolysis products (such as C_2H_3) is very low, and the concentration of both is only 0.01 times that of the Ske61 mechanism. The results showed that the Z77 mechanism was weakened at a large strain rate. It could not

accumulate enough free radicals and small molecular hydrocarbon products to release heat in the oxidation process, so it was blowoff in the simulation of the $M = 0.1$ condition. Compared with ARC mechanisms, WD2 and JL4_{kero} can simplify the pyrolysis process of fuel, and the macromolecular fuel is directly decomposed into CO , H_2O , and CO_2 , and heat is released. Therefore, the influence of flow is small at the high strain rate, and the combustion temperature and heat release rate are maintained.

However, under the $M = 0.2$ condition, the temperature rise of the four unstable combustion mechanisms is slight, and only the JL4_{kero} mechanism can produce a high heat release rate and increase the temperature. The too-low reaction-heat release rate cannot provide enough heat to stabilize combustion through expansion, making vortex street flow more and more intense, so the return zone is shortened. Finally, the flame is broken and extinguished.

This indicates that different chemical mechanisms will change the distribution of components in the flow field, resulting in different local reactions and heat release. At the same time, the high flow strain rate K_s will destroy the reaction layer of combustion, and reduce the generation of intermediate products and free radicals of kerosene macromolecular cracking, so that the subsequent heat release reaction cannot be carried out due to the lack of reactants, resulting in the blowoff. Especially for the ARC mechanisms that preserve the pyrolysis of kerosene into small molecule hydrocarbon products, the high flow strain rate K_s weakens the pyrolysis process and reduces the molar concentration of the pyrolysis products. WD2 mechanism of the pyrolysis of kerosene does not depend on small molecule hydrocarbon products, intermediate CO can be obtained directly but does not consider the H_2 generation and oxidation,

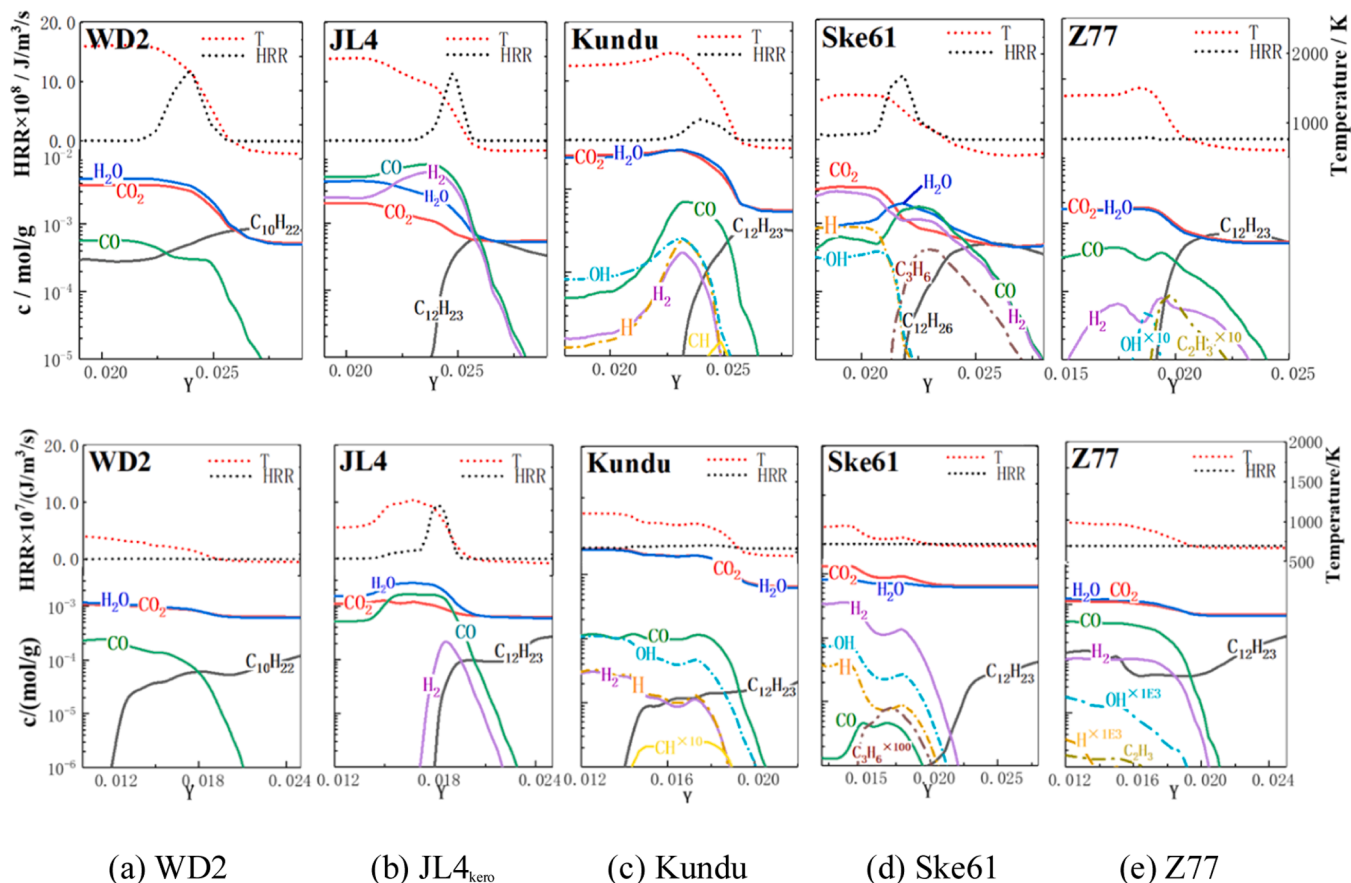


Fig. 24. Distribution of heat release rate, temperature and molar concentration of components (above: M0.1 condition, below: M0.2 condition).

the reaction process is too simple, can affect the combustion heat release properties, lower the total heat release rate. Therefore, blowoff also appears in the simulation of M0.2 working conditions.

In the M0.2 condition, the Kundu mechanism takes CH as an example, the highest concentration of Kundu is only 0.01 times that of M0.1; The Ske61 mechanism takes C_3H_6 as an example, and the concentration is 1×10^{-4} times that in M0.1 condition; Z77 mechanism takes C_2H_3 as an example, the concentration is reduced to 15 % of M0.1. In addition, free radicals H and OH are also relatively reduced. The reduction of these highly active components further reduces the generation of intermediate CO and H_2 , which makes the subsequent exothermic reaction unable to proceed, and the instantaneous heat release rate is low.

In summary, a high strain rate will weaken the fuel pyrolysis process of the ARC mechanisms, reduce free radicals and small molecular hydrocarbon products, thus destroying the combustion reaction, characterize the unstable combustion region of the flow field, and can be used to judge the local blowoff. Therefore, the blowoff limit of the K_s combined mechanism can be used to judge the local combustion state and predict blowoff.

6. Conclusions

This paper uses various chemical reaction mechanisms to study the influence of mechanisms on near the extinction limit. The following conclusions are drawn in the summary of the whole paper.

- (1) In the two-phase V-gutter flame holder simulation, the effect of kerosene mechanisms on flame stability was analyzed: the overall WD2 and ARC-type Kundu, Ske61, and Z77 mechanisms extinguish at M0.2 condition, and Z77 failed to burn at M0.1

condition. The heat release rate of each mechanism is extremely low in these cases. The strong turbulence will weaken the reaction and destroy the pyrolysis process of ARC-type mechanisms, reducing the formation of pyrolysis products such as small molecule hydrocarbon products and free radicals. As a result, the ARC-type mechanisms are insufficient to support the subsequent oxidation reaction due to too few pyrolysis products during blowoff, and the heat release rate is low. However, because H_2 is not considered, the reaction is too simple, and the heat release is low, the overall WD2 also shows a blowoff phenomenon under the M0.2 condition. These phenomena show that mechanism selection greatly influences bluff body blowoff simulation.

- (2) The simplified mechanism of accurate ignition delay time and flame propagation velocity may not apply to bluff body combustion, so the proper mechanism should be selected to get correct results in CFD simulation.
- (3) In the simulation of bluff body flame, the species and reactions of JL4 are the simplest. Due to the lack of suitable intermediate products, the heat release rate is high, so it is impossible to simulate the state of blowoff.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Data availability statement

The data that supports the findings of this study are available from the corresponding author, Fang Wang., upon reasonable request.

CRediT authorship contribution statement

Fang Wang: Writing – review & editing, Project administration, Methodology. **Yuhan Gao:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Shuo Shen:** Visualization, Validation, Resources, Formal analysis, Data curation, Conceptualization. **Yudong Wang:** Supervision, Software, Methodology. **Li Dou:** Software, Methodology. **Jie Jin:** Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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